

## RELATIVISTIC ADDITION OF VELOCITIES

No two objects can have a relative velocity greater than  $c$ .

e.g. A spacecraft moving at  $0.9c$  fires a missile moving at  $0.8c$  relative to it.

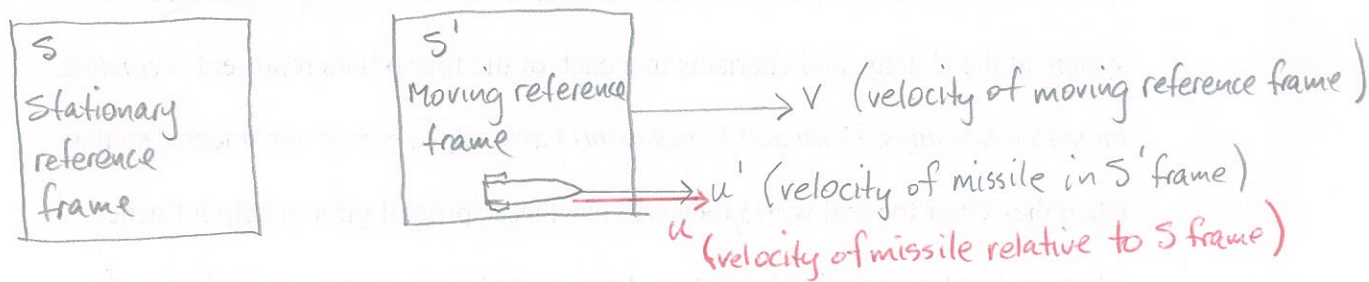
A stationary observer sees this as  $1.7c$ , which is impossible.

Velocities must transform according to the Lorentz transformation, leading to a non-intuitive result called Einstein velocity addition.

( See Ch 8.4 - Relativistic addition of velocities )

Solution to the "problem"

Consider two observers in relative motion to each other and both are observing the motion of the missile.



The equation describing  $u$  is given by:

$$u = \frac{u' + v}{1 + \frac{u'v}{c^2}}$$

where

- $u$  = velocity of the moving object in the stationary frame.
- $u'$  = velocity of the moving object in the moving frame. ( $= 0.8c$ )
- $v$  = velocity of the moving frame of reference ( $= 0.9c$ )
- $c$  = speed of light.

$$\begin{aligned}
 10. \quad u &= \frac{u' + v}{1 + \frac{u'v}{c^2}} \\
 &= \frac{0.8c + 0.9c}{1 + \frac{(0.8c)(0.9c)}{c^2}} \\
 &= \frac{1.7c}{1.72} \\
 &= \underline{0.988c}
 \end{aligned}$$

### EXAMPLE

Two spaceships are moving apart from each other, A at  $0.7c$  and B at  $0.5c$  in the other direction. What is the velocity of A as seen by an observer in B?

Consider B as the stationary reference frame.

Take the direction of B as +ve.

$$v = -0.7c$$

$$u = 0.5c$$

$u' =$  velocity of A relative to B

$$\begin{aligned}
 u' &= \frac{u - v}{1 - \frac{uv}{c^2}} \\
 &= \frac{0.5c - (-0.7c)}{1 - \frac{(0.5c)(-0.7c)}{c^2}} \\
 &= \frac{1.2c}{1.35} \\
 &= \underline{0.889c}
 \end{aligned}$$

∴ Spaceship A is receding at  $0.889c$  from B.